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Semi-empirical dipole moment of carbon monoxide and line lists for all its isotopologues revisited

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ABSTRACT

The semi-empirical permanent electric dipole-moment function (DMF) for the ground state of the CO molecule has been reconstructed analytically in the entire range of the inter-nuclear distance $r \in [0, +\infty)$ by means of the simultaneous non-linear least-squares fitting (NLLSF) of the selected experimental intensities for the main $^{12}C^{16}O$ isotopologue (including those with sub-percent uncertainties) and the ab initio permanent dipole moment. The ab initio DMFs were evaluated using single-reference coupled cluster (SR-CCSD(T)) and multi-reference averaged coupled-pair functional (MR-ACPF) methods. The ab initio data were involved in the NLLSF procedure to propagating smoothly the semi-empirical DMF outside the local region covered by the experimental intensities for the lowest vibrational $v' \leq 11$ levels. The derived mass-invariant DMF possesses the physically correct asymptotic behavior in both the united-atom and dissociation limits as well as reproduces the vast majority of the measured intensities in $v'' = 0 \to v' \in [0, 6]$ and $v'' = 1 \to v' = 4, 5$ bands within their experimental uncertainties. The resulting DMF and the mass-corrected potential-energy function of Meshkov et al. (2018)[16] were used to upgrade line lists for all CO isotopologues in the wide range of vibrational and rotational quantum numbers $v \in [0, 41]$, $\Delta v \in [0, 6]$, $J \in [0, 150]$ ($J \in [0, 200]$ for the 0-0 and 1-0 bands). The predicted intensities are compared with their experimental counterparts, which were not involved in the present NLLSF, to highlight presumable random and systematic errors in the measured data; in particular, the intensities in the “abnormal” $0 \to 5$ band of the $^{12}C^{16}O$ isotopologue and some other bands of minor isotopologues should be revisited experimentally. The resultant line list should be considered superior to previous efforts in terms of accuracy.

1. Introduction

Knowledge of accurate reference spectroscopic parameters for the gaseous carbon monoxide (CO) is essential for many applications. It is a pollutant in the terrestrial atmosphere (especially in the troposphere), occurring from both natural (i.e., biomass burning) and anthropogenic (i.e., automobile exhausts) sources [1]. It is routinely monitored using different spectrometers onboard remote-sensing satellites, including ACE [2], and TROPOMI [3]. These remote sensing missions rely on the reference spectroscopic data provided in the HITRAN molecular spectroscopic database [4]. Spectral lines of carbon monoxide have been observed in the atmospheres of Mars [5] and Venus [6] and are now routinely identified in exoplanets using the cross-correlation method (see, for instance [7]). CO is present in solar [8] and stellar [9] atmospheres, and therefore high-temperature line lists (i.e., those that contain a larger amount of vibrational and rotational levels) of CO for different isotopologues are required. Studies of the combustion processes also rely on this information [10]. For these purposes, the HITRAN group also maintains the HITEMP database [11], which follows the same formalism as HITRAN but contains substantially more transitions.

Due to its importance, there were many experimental and theoretical works devoted to the spectroscopy of CO in the last hundred years. Despite being a simple diatomic many controversies do exist between different data sources especially for intensities at a sub-percent level which is required for many applications. In 2015, in order to obtain a consistent data set for all isotopologues of CO in HITRAN and HITEMP, Li et al. [12] have developed new semi-empirical piece-wise dipole moment function (DMF) using a direct-fit approach to the selected experimental data and ab initio calculations carried out for this purposes. Combining this DMF

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with the empirical potential energy function (PEF) from Coxon and Hajigeorgiou [13], Li et al. [12] calculated a new line list for six stable and three radioactive isotopologues of CO. This new line list proved to be a clear improvement over previous compilations and has been adapted not just for HITRAN and HITEMP but also for ExoMol [14] and GEISA [15] databases. Nevertheless a number of issues (and potential issues) have been identified with Li et al. [12] line list, part of which have been fixed when adapted into HITRAN and HITEMP, yet other issues remained open and are now addressed in this work.

- As it was mentioned in the Li et al. [12] paper the LEVEL program had some incompatibilities with the PEF from Coxon and Hajigeorgiou [13] which resulted in line position deviations that grew slowly with the rotational quantum numbers. After the publication of Li et al. [12] work the compatibility issue was solved and the line list that currently resides on HITEMP website contains recalculated line positions. These improved line positions were also taken for the HITRAN2016 edition, although many of them were updated with high quality experimental measurements especially those obtained with a frequency comb method. In this work we calculated the line positions using the PEF from Meshkov et al. [16] and the comparison with best experimental data shows its high quality.
- In Li et al. [12] it was also recognized that the quality of the intensities of high-overtone transitions suffers from different numerical issues. Therefore the line list was limited to $\Delta v = 13$. Later on Medvedev et al. [17,18] have shown that the intensities of high-overtone transitions in Li et al. [12] (and many other line lists calculated in a similar way) suffer from the use of the naturale cubic splines when interpolating ab initio points in the DMF, and to lesser extent use of only double precision in the calculations.[1] It was shown that the intensities from Li et al. are only reliable to $\Delta v = 7$. It was demonstrated in Medvedev et al. [19] that the best way to interpolate ab initio points was to fit them to analytical function. Moreover, Medvedev and Ushakov [20–22] have determined that the rotational distribution in the intensities of high overtones is also very sensitive to the choice of such analytical functions.
- After the release of HITRAN2016 a number of new intensity measurements with reportedly low uncertainties in different bands have emerged. The new studies of the first [23] and third overtone [24,25] and the corresponding hot bands have agreed well within mutual uncertainties with the intensities in the HITRAN2016 database. However, new highly accurate measurements of the lines of principal isotopologue of CO in the fundamental band [26] have suggested that the intensities in HITRAN2016 for this band are overestimated by about 2%. In contrast recent measurements [27–29] in the second overtone (3-0 band) suggest that the HITRAN2016 intensities in this band need to be increased by 2.6%. For the new edition of the HITRAN database (HITRAN2020 [30]) a temporary solution of scaling of the intensities of these bands have been applied. However, a proper revision of the dipole-moment function is required to obtain better intensities for the hot bands and isotopologues.
- In Li et al. [12] the same DMF was applied to calculate intensities of all isotopologues which should yield same level of accuracy for all the isotopologues assuming that there is no Born–Oppenheimer breakdown (BOB), which is traditionally considered to be a very reasonable assumption for the ground state of the CO molecule. Nevertheless, as it was noted in Li et al. intensities of different isotopologues in calculated line list differed from the ones measured in the experiments available at the time. The deviations were of different signs ranging in magnitude from 0 to 10%. They were predominantly systematic within a band but differed from study-to-study and band-to-band of particular isotopologues. Li et al. have discussed the possibility that since the experimental studies did not employ mass-spectrometers to monitor abundance of isotopologues in their spectra, it would have been possible that it was quite different from the "natural" abundance (calculated from atomic abundances from [31]) provided in the HITRAN database. New measurements [24–26,28,29], which also did not use mass spectrometers, did not clarify but only added further confusions in the discrepancies. In order to model experimental intensities per isotopologue Perevalov and Karlovets [32] fitted experimental data and created an empirical model for calculating intensities of individual isotopologues.Thus, we have recalculated the intensities of CO isotopologues using the new PEF from Meshkov et al. [16] and the new DMF described in this work. We also tested different programs for calculating the line lists. By doing so, we have investigated if the isotopic deviations mentioned above are not caused by any artifacts of the PEF, DMF, and associated programs used in the Li et al. study.

2. The semi-empirical dipole moment construction

2.1. The regularization procedure for the DMF non-linear least-squares fitting

The semi-empirical permanent dipole moment function, $d^{se}(r)$, has been determined for the ground state of CO molecule by means of the iterative minimization of the non-linear functional

$$\chi^2 = \chi^2_{\rm emp} + w\chi^2_{\rm ab}, \tag{1}$$

where the weighting factor w has been monotonically decreased from 1 to an appropriate small value, $w = 0.02$, until the $\chi^2_{\rm emp}$ value,

$$\chi^2_{\rm emp} = \sum_{k=1}^{N_{\rm emp}} \left(\frac{{\rm TDM}^{\rm se}_k - {\rm TDM}^{\rm emp}_k}{\sigma^{\rm emp}_k}\right)^2, \tag{2}$$

becomes almost insensitive to the $\chi^2_{\rm ab}$ value,

$$\chi^2_{\rm ab} = \sum_{i=1}^{N_{\rm ab}} \left(\frac{d^{se}(r_i) - d^{\rm ab}(r_i)}{\sigma^{\rm ab}_i}\right)^2. \tag{3}$$

Hereafter, $N_{\rm emp}$ is the total number of empirical transition dipole moments, ${\rm TDM}^{\rm emp}_k$, which were extracted from the intensity measurements in Section 2.3, and, then, involved in the non-linear least-squared fitting (NLLSF) procedure, together with the corresponding uncertainties $\sigma^{\rm emp}_k$. $N_{\rm ab}$ is the number of *ab initio* DMF values defined in the point-wise form, $d^{\rm ab}(r_i)$, and $\sigma^{\rm ab}_i$ are their uncertainties. The ab initio data was included in the NLLSF procedure (1) in order to constrain and properly propagate the $d^{se}(r)$ function outside the experimental intensity region, which is restricted by the $v'' = 0 \to v' \in [0, 6]$ and $v'' = 1 \to v' = 4, 5$ vibrational bands (see Section 2.3).

The $d^{se}(r)$ function was approximated within the entire range of inter-nuclear distance $0 \le r < \infty$ by the analytical formula

$$d^{se}(r) = (1 - e^{-a_1 r})^7 r^{-4} \sum_{k=0}^{N} b_k T_k(z(r)), \tag{4}$$

where $T_k(z)$ are Chebyshev polynomials of the first kind and b_k are the corresponding expansion coefficients. The mapping function used in Eq. (4),

$$z(r) = 1 - 2e^{-a_0 r}\left(1 + a_0 r + \frac{1}{2}a_0^2 r^2\right), \tag{5}$$

[1] The double precision is sufficient for $\Delta v < 20$.

transfers the conventional radial coordinate defined on infinite interval $r \in [0, +\infty)$ onto the required finite domain $z \in [-1, +1)$ providing additionally the high density of the r-grid in the vicinity of the equilibrium distance, r_e, where the wavefunctions of the lowest vibrational levels studied are localized (see Appendix A).

The $d^{se}(r)$ function (4) satisfies the physically justified asymptotic behavior in both the united-atom (UA) and dissociation limits [33–37]:

$$d^{se}(r \to 0) = A_3 r^3; \quad d^{se}(r \to +\infty) = d_4 r^{-4}, \tag{6}$$

where the short-range A_3 and long-range d_4 coefficients obey the relations:

$$A_3 = (a_1)^7 \sum_{k=0}^{N} (-1)^k b_k; \quad d_4 = \sum_{k=0}^{N} b_k. \tag{7}$$

The semi-empirical kth $(v', J') \to (v'', J'')$ transition dipole moments were calculated as

$$\mathrm{TDM}_k^{se} = \int_0^\infty \psi_{v'J'}(r) d^{se}(r) \psi_{v''J''}(r) dr, \tag{8}$$

where $\psi_{vJ}(r)$ are properly normalized ro-vibrational wavefunctions,

$$\int_0^\infty |\psi_{vJ}(r)|^2 dr = 1.$$

The required eigenvalues and eigenfunctions were obtained for each ro-vibrational bound level, (v, J), of the ground CO state by means of a numerical solution of the radial equation:

$$-\frac{\hbar^2}{2\mu_i}\frac{d^2}{dr^2}\psi_{vJ}(r) + \left[U_i^{eff}(r) + B_i(r)[1 + q_i(r)]Y - E_{vJ}\right]\psi_{vJ}(r) = 0,$$

$$B_i(r) = \frac{\hbar^2}{2\mu_i r^2}; \qquad Y \equiv J(J+1) \tag{9}$$

with the effective mass-dependent interatomic potential $U_i^{eff}(r)$ involving rotationless Born–Oppenheimer (BOB) correction to the BO potential $U^{BO}(r)$

$$U_i^{eff}(r) = U^{BO}(r) + \frac{m_e}{\mu_i}[u(r) + \gamma_i \delta u(r)]; \quad u \gg \delta u \tag{10}$$

and the modified centrifugal term $B_i(r)[1 + q_i(r)]Y$, where

$$q_i(r) = \frac{m_e}{\mu_i}[q(r) + \gamma_i \delta q(r)]; \quad q \gg \delta q. \tag{11}$$

Here $\mu_i = M_i^C M_i^O/(M_i^C + M_i^O)$ is the reduced molecular mass for the particular isotopologue i, and dimensionless parameter $\gamma_i = (M_i^O - M_i^C)/(M_i^C + M_i^O)$.

Both U^{BO} potential and correction functions ($u, \delta u$ and $q, \delta q$) were semi-empirically determined in our preceding work [16] in the framework of LeRoy's direct-potential-fit (DPF) procedure based on the exhaustive set of the highly accurate experimental line positions assigning to the ro-vibrational ground state levels $v' \in [0, 41], J' \in [0, 133]$ of all stable isotopologues. The effective potential (10) defined analytically in the entire r-range provides unprecedentedly high accuracy estimates of the ro-vibrational term values for the ground CO state. Indeed, the 3-0 band lines measured recently for the main CO isotopologue [27,38,39] are reproduced by our potential with the absolute systematic error is about 30–80 kHz only, as can be seen in Fig. 1.

To probe regular perturbations effects on the ro-vibrational wavefunctions used for the TDM_k^{se} calculation we have solved the radial Eq. (9) by removing subsequently J-invariant (u, and δu) and J-dependent (q, and δq) BOB correction functions from the effective Hamiltonian. The resulting "deperturbed" ro-vibrational eigenfunctions were then employed in calculations of the TDM_k^{se} values and in the refitting of the semi-empirical DMF. The modified $d^{se}(r)$ function is found to almost exactly coincide with the initial DMF in

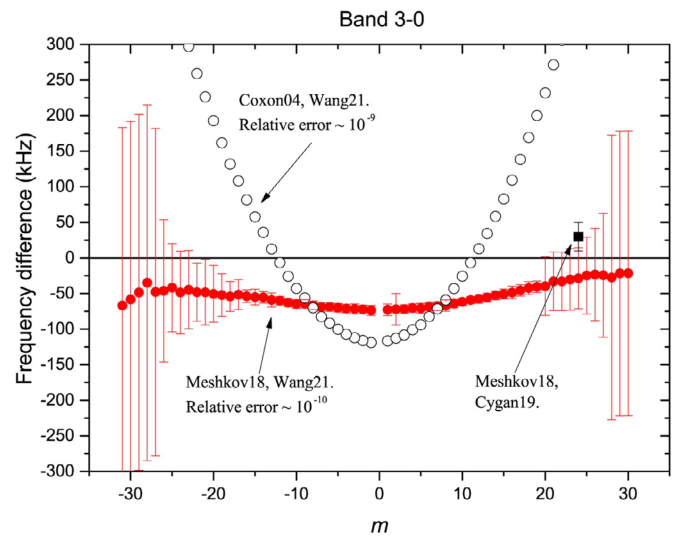

Fig. 1. The difference between highly accurate experimental line positions recently measured [27,39] for the $0 \to 3$ absorption band of $^{12}C^{16}O$ isotoplogue and their theoretical counterparts evaluated with the semi-empirical ground state PEFs of Meshkov [16] and Coxon [13], respectively.

the case where only $u(r)$ and $\delta u(r)$ functions were removed from the effective inter-atomic potential (10). A similar result was already obtained in Ref. [18], where it was shown that small changes in the rotationless potential do not affect the transition intensities, excluding those at the anomalies. Exclusion of the rotationally dependent part of the BOB correction from the radial Eq. (9) also leads to a very small variation of the $d^{se}(r)$ function while the corresponding χ^2_{emp} value becomes even slightly less than those of the full effective Hamiltonian approximation.

It should be noted that the semi-empirical matrix elements, TDM_k^{se}, included in the regularization NLLSF procedure (1) are sensitive to the semi-empirical DMF only in a narrow range of r, which is approximately determined by the positions of the turning points for the highest vibrational level involved in the fit. In our case this is $v' = 11$ level (see Table 1) and, hence, $r \in [0.96, 1.44]$ Å [40]. Outside this region, the $d^{se}(r)$ function mainly depends on the ab initio data only.

2.2. Ab initio calculation of the DMF

The ab initio DMF calculations were started with applying the so-called "gold standard" of the modern quantum-chemistry method - the spin-restricted close-shell coupled cluster approach, with the perturbative treatment of triple excitation [CCSD(T)] using the full electrons atomic basis sets of the aug-cc-pCVnZ ($n = 5, 6$) quality. The single-reference CCSD(T) calculations were performed using the standard "frozen core" approximation and the explicit correlation of all electrons of the molecule. The d^{ab} values were evaluated by the finite-field (FF) technique on the restricted interval, $r \in [0.4, 1.7]$ Å. All CCSD(T) calculations were carried out with the GAMESS-US program package [41].

To extend the ab initio DMF function in a large distance, the multi-reference configuration interaction (combined with the Davidson correction during the FF calculation) and averaged coupled-pair functional methods were implemented. We performed MR-CI/MR-ACPF calculations with the same reference space as it was used in Refs. [12,42], namely, the molecular orbitals (MO) $1-3\sigma$ were doubly occupied in all configurations, while the remaining eight electrons were distributed over the $4-6\sigma$, $1-2p_x$, and $1-2p_y$ orbitals. The optimized MOs were obtained using the

Table 1
The experimental absolute intensity data, S_k^{exp}, collected for the $v''=0 \to v'=$ 0-6 and $v''=1 \to v'=4,5$ ro-vibrational transitions in the ground $X^1\Sigma^+$ state of the principal $^{12}C^{16}O$ isotopologue. ‡ - the data has been previously used in Ref. [12].

$v'-v''$	lines	References	Comments
0-0	R2-R21	‡Aenchbacher10 [52]	40% lines were used
0-0	R7-R23	‡Birk96 [50]	two from four sets was used
1-0	P4	‡Hartmann86 [53]	not used
1-0	P30-R30	Devi18 [26]	70% lines were used (high J-values were exculuded)
1-0	P23-R25	‡Zou02 [54]	80% lines were used
2-0	P24-R23	‡Zou02 [54]	not used
2-0	P29-R30	‡Devi12 [55]	85% lines were used
2-0	P25-R28	Loos17 [23]	about 40% lines were used
2-0	P24-R22	Esteki17 [56]	not used
3-0	P26-P28	Reed17 [29]	all 3 lines were used
3-0	P27,P28	‡Wojtewicz13 [57]	not used
3-0	P18-R21	‡Chackerian01 [58]	not used
3-0	P20-R20	‡Jacquemart01 [59]	not used
3-0	P25-R25	‡Sung04 [60]	80% lines were used
3-0	R23	Cygan19 [27]	1 line with 0.07% error was used
3-0	P39-R42	Borkov20 [24]	80% lines were used
4-0	P34-P19	Campargue15 [61]	70% lines were used
4-0	P34-R38	Bordet20 [25]	80% lines were used
4-0	P16-R23	‡Chackerian76 [62]	not used
4-0	P30-R31	Borkov19 [28]	80% lines were used
4-1	P15	‡Wojtewicz13 [57]	1 from 2 lines was used
4-1	P25-R20	Borkov20 [24]	20% lines were used
5-0	P3-18	‡Chung05 [63]	all used with 25% uncertainty
5-1	P6,11	Campargue15 [61]	all 2 lines were used
5-1	P18-R20	Campargue20 [61]	15% lines were used
6-0	P20-R11	Tan17 [64]	70% lines were used
5-6,7-8,8-9,10-11	P8-13	Weisbach73 [65]	80% lines were used

multi-configuration SCF (MC-SCF) method realized in the same active space.

The MCSCF calculation of the CO molecule in the region $r > 5.5$ Å becomes not trivial since there are $(1,2)\Sigma^+$ and $(1)\Delta$ states which belong to the same a_1 irreducible representation of the C_{2v} point group and asymptotically converge to the ground states of C and O atoms. This leads to a strong admixture of different MOs at a large distance where the corresponding molecular states become degenerate. It has already been shown in Refs. [37,43,44] that the state-averaged MCSCF with proper weights of the states can produce the correct long-range asymptote of internuclear potential and DMF function. In order to obtain the MOs which give the correct long-range behavior, one should use the state-averaged MCSCF calculation on the singlet states $(1,2)\Sigma^+$ and $(1)\Delta$ with equal weights of the states.

On the other hand, at short and intermediate distances, where the energy gap between the ground and excited states becomes larger, MCSCF with equal weights is not required anymore, and in practice, it leads to convergence issues at distances near the equilibrium distance r_e of the ground state. Hence, it is necessary to gradually increase the weight of the X-state relative to the $(2)^1\Sigma^+$ and $(1)^1\Delta$ states from 1 at the long-range region to a dominant value near r_e. It has been done by means of the dynamic weighting $w_{X^1\Sigma^+} = 1 + 120/r^5$ of the ground state while the weights of the $(2)^1\Sigma^+$ and $(1)^1\Delta$ states were kept to be 1 at all distances. This weighting was determined to give the maximal weight to the ground state in the short-range region while not impairing the DMF's long-range behavior.

The dynamic weighting did not completely remove discontinuities and incorrect overall behavior in the resulting $d^{ab}(r)$ functions calculated by the MR-CI/MR-ACPF methods at large distances. It was established [37] that these discontinuities are due to the presence of "undesirable" configurations in the reference space used. To trace down the problem, all configurations with the CI coefficients less than 0.01 have been automatically removed. Then, we examined the remaining configurations at each distance and have identified three minor ones whose excluding from the active space greatly improved the long-range tail of the $d^{ab}(r)$ function.

The final d^{ab}-values were evaluated using both the FF and expectation-value techniques. Although the FF method is more commonly used for the DMF calculation of the CO molecule [12,42], we found that in the long-range region, the expectation-value technique provides much more stable results, especially when the dynamic weighting was used.

A minor scalar-relativistic effect was taken into account in the present ACPF and MRCI calculations by means of the effective Douglas–Kroll–Hess (DKH) Hamiltonian and the aug-cc-pwCVnZ-DK($n = 3$–5) atomic basis sets used [45]. Both MCSCF and ACPF/MRCI calculations were conducted owing to the MOLPRO 2010.1 package [46].

2.3. The input data on experimental intensities

The necessary input empirical transition dipole moments, TDM_k^{emp}, accompanied by their uncertainties, σ_k^{emp} in Eq. (2), were extracted from the corresponding experimental transition intensities, S_k^{exp}, measured at or recalculated to the standard temperature, $T_0 = 296$ K, by the formula

$$S_k = c_1 \frac{g_i g_s I_a \mathrm{HL}_k \tilde{\nu}_k}{Q_{tot}} \left(1 - e^{-c_2 \tilde{\nu}_k}\right) e^{-c_2 E_{v''J''}} |\mathrm{TDM}_k|^2, \tag{12}$$

$$c_1 = \frac{1}{4\pi\epsilon_0} \frac{8\pi^3}{3hc} = 4.16237903 \times 10^{-19}\ \mathrm{cm^2 D^{-2}}, \tag{13}$$

$$c_2 = \frac{hc}{k_B T_0} = 4.8607343 \times 10^{-3}\ \mathrm{cm}, \tag{14}$$

which is combined from the equations given in Refs. [47,48]. Here, g_i and g_s are the state-independent (nuclear) and state-dependent statistical weights, see Table 1 in Šimečková et al. [48]; I_a is abundance; $\tilde{\nu}_k = \nu_k/c$ is the kth transition wave number in cm^{-1}; $\mathrm{HL}_k \equiv |m|$ is the Hönl–London factor, equal to $J'+1$ for the P-branch and J' for the R-branch; Q_{tot} is the total statistical sum (including nuclear weights) at T_0; $E_{v''J''}$ is the lower-state energy in cm^{-1}; and TDM$_k$ is the transition moment in D. The dimension of S_k is cm/molecule. The statistical sums for all isotopologues needed for calculations of the intensities at arbitrary temperatures are given in the supplementary file **Q.rar.**

The sources of the experimental data considered are listed in Table 1. Besides the absolute absorption intensities belonging to $v'' = 0 \to v' \in [0, 6]$ and $v'' = 1 \to v' \in [4, 5]$ ro-vibrational transitions of the principal isotopologue, we have included in the NLLSF procedure the permanent dipole moments obtained from the Stark splitting measurements of the 0-0 transition [47,49,50]. At the same time, we had to exclude from the fit the experimental intensity ratios corresponding to the emission transition from the higher excited vibrational levels $v' \le 38$ of the ground state [51] since these data was found to contradict the directly-measured absorption intensities used above. It should be noticed that we have particularly used in the regularization NLLSF procedure (1) only the empirical TDM values corresponding to the intensities measured in the most abundant $^{12}C^{16}O$ isotopologue. The intensities of the most abundant isotopologue are easier to measure accurately due to high signal-to-noise ratio in “natural” samples, and assuming that there is no BOB this should be sufficient to construct the DMF suitable for all isotopologues. This also allows to highlight presumable random and systematic errors in the empirical data of minor isotopologues.

2.4. Line lists generation for all CO isotopologues

We used the resulting semi-empirical $d^{se}(r)$ function and the highly accurate semi-empirical PEF [16] to generate new line lists for all CO isotopologues in the diapason of vibrational and rotational quantum numbers $v \in [0, 41]$, $\Delta v \in [0, 6]$, $J \in [0, 150]$ ($J \in [0, 200]$ for the 0-0 and 1-0 bands).

The Einstein A coefficient (in s^{-1}) was calculated for the kth transition by the formula

$$A_k = c_3 \tilde{\nu}_k^3 \frac{\mathrm{HL}_k}{2J'+1} |\mathrm{TDM}_k|^2, \tag{15}$$

$$c_3 = \frac{1}{4\pi\epsilon_0}\frac{64\pi^4}{3h} = 3.1361887 \times 10^{-7}\ \mathrm{cm}^3\ \mathrm{D}^{-2}\mathrm{s}^{-1} \tag{16}$$

and the respective transition intensity by Eq. (12) with abundances equal to unity for all isotopologues.

3. Results and discussion

3.1. The resulting approximation of experimental intensities

The representation of experimental intensities is characterized by the data in Table 2 where relative differences between the calculated and measured line intensities averaged over the bands are shown. Nearly for all bands, they are about a few percent, and all are less than the experimental uncertainties.

3.2. The ab initio DMFs

The resulting ab initio DMFs obtained in the framework of the SR-CCSD(T), MR-CI and MR-ACPF calculations are depicted in Fig. 2. The corresponding ab initio point-wise functions obtained by all methods used are tabulated in Supplementary Material as well. In the present work, the sign of the dipole moment for the ground CO state is defined to be positive for the polarity C^+O^- .

All ab initio calculations unambiguously confirm an additional change of the sign of DMF in a range of small inter-nuclear separations, as it has been predicted in Ref. [35] using the UA theory.

Table 2
Relative error of the representation (in %) of the experimental intensities (averaged over the rotational distribution) by using of either the semi-empirical (SE) or the ab initio DMFs. Both CCSD(T) and ACPF point-wise functions were approximated by the ordinary polynomial function of 4th order.

Band	No. of lines	SE	CCSD(T)	ACPF	Exp. uncertainty, %
0-0	34	0.8	6.0	8.0	4.7
1-0	72	0.8	4.4	2.5	1.7
2-0	84	0.1	2.4	1.7	0.3
3-0	89	1.1	6.5	1.4	2.4
4-0	138	0.9	32	36	2.1
4-1	26	2.5	8.8	2.5	6.6
5-0	16	22	142	92	25
5-1	20	3.4	29	35	10
5-4	2	0.5	3.8	1.8	2.5
6-0	20	1.8	42	2.0	3.6
6-5	3	0.7	4.3	2.3	1.8
11-10	1	2	6.4	4.6	2.9

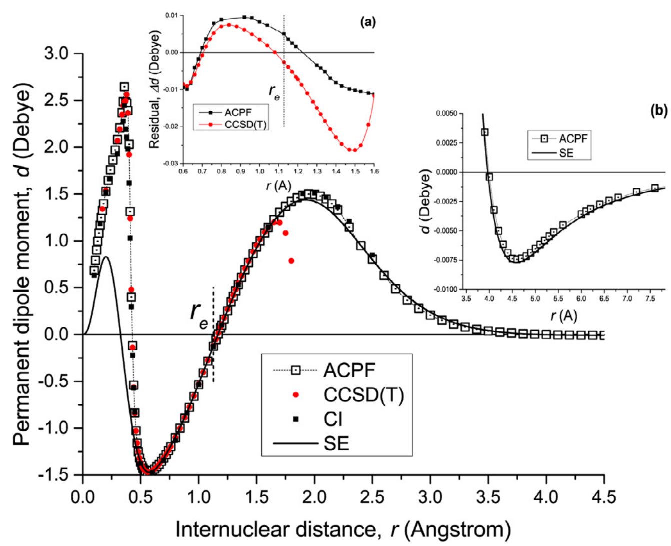

Fig. 2. The semi-empirical DMF (solid line) and ab initio data (points) calculated by the SR-CCSD(T), MR-CI and MR-ACPF methods. Inset (a) demonstrates the difference between the semi-empirical DMF and its ab initio CCSD(T) and ACPF counterparts, with randomly scattered ACPF points being removed from the plot. Inset (b) is the zoomed long-range tail of the semi-empirical and ACPF DMFs. r_e is the equilibrium bond length.

Thus, in the entire range of the inter-nuclear distance $r \in [0, +\infty)$, the permanent electric dipole moment for the ground state of the CO molecule changes its sign three times at r(Å) $\approx$ 0.4, 1.14, and 4.0, respectively.

The single-reference approximation breakdown in the CCSD(T) method leaded to the nonphysical r-behavior of the $d^{\mathrm{CCSD(T)}}(r)$ function at r(Å) $>$ 1.6, that followed by incorrect slope of the corresponding CCSD(T) ground state PEF [66]. All ab initio methods demonstrated an irregular behavior of the resulting DMF points at a very small distance r(Å) $<$ 0.4. This drawback could be attributed to a limitation of the conventional Gaussian atomic basis sets to simulate the electronic density near the nuclei as well as to the linear dependence of the basis set functions at small r-distances.

The $d^{\mathrm{ACPF}}(r)$ function coincides with its semi-empirical counterpart $d^{se}(r)$ (see the Inset (a) on Fig. 2) within $\pm$0.01 D at least in the interval $r \in [0.6, 1.6]$ Å, where the lowest vibrational wavefunctions are localized. Nevertheless, this small (from the state-of-art computational standard viewpoints) absolute systematic error in the ACPF DMF still does not allow to reproduce (predict) the most part of the empirical TDM$_k^{\mathrm{emp}}$ values with the required

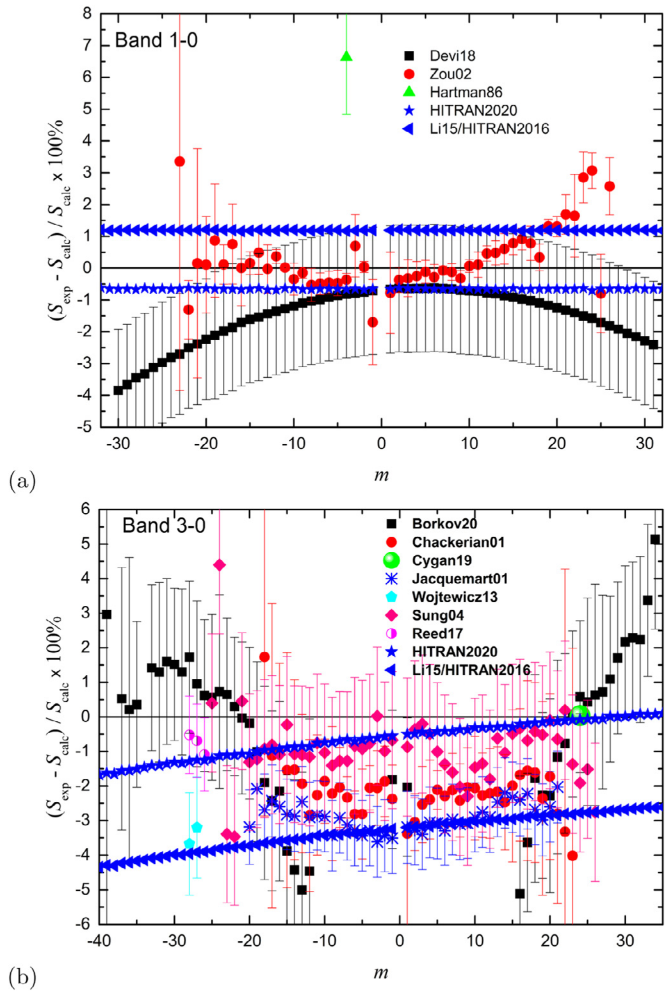

Fig. 3. Comparison of line intensities of $^{12}C^{16}O$ fundamental (a) and second overtone (b) bands with the present semi-empirical study as a reference. Note the different scales in the panels of the plot. "Calc" refers to semi-empirical results from the present study, while "exp" refers to the experimental studies from Refs. [26,53,54] for the 1-0 band, Refs. [24,27,29,57,59,60] for the 3-0 band, or two last editions of HITRAN [4,30]. Note that for HITRAN2020 intensities from Li et al. [12]/HITRAN2016 were scaled to agree better with experiments from Refs. [26,27,29].

spectroscopic accuracy (about 1%). Furthermore, our additional attempts to diminish the basis-set incompleteness error (including extrapolation to the complete basis set) failed to improve the ab initio results. The present comparison of the ab initio DMFs with their semi-empirical counterpart leads to the statement that the systematic diminishing the basis set incompleteness error does not guarantee at all the expected decrease of the final error in DMFs.

The further refinement of the ab initio DMFs can be apparently associated with a systematic decrease of the electron-correlation error [67]. In principle, it could be done by accounting in the coupled-cluster method for higher (triple and quadruple) excitation in explicit (non-perturbative) form. However, this is a very expensive computational task, especially in combination with the required high-quality atomic basis set.

3.3. The semi-empirical DMF

The adjusted parameters of the resulting mass-invariant semi-empirical DMF obtained for the ground state of CO molecule are given in Table 3. The non-linear fitting parameter a_0 corresponds

Table 3
The fitted parameters of the mass-invariant semi-empirical DMF defined by Eq. (4) on $r \in [0, +\infty)$ Å for the CO ground state. Two non-linear parameters $a_{0,1}$ are in 1/Å , the linear expansion coefficients b_i in DÅ^4. The FORTRAN subroutine for the present DMF is given in the Supplementary Material.

a_0	1.88122
a_1	2.62390
b_0	6.94695
b_1	8.22984
b_2	−3.76974
b_3	−10.57303
b_4	−7.12739
b_5	−2.35278
b_6	0.88068
b_7	1.27112
b_8	0.95583
b_9	0.33660
b_{10}	0.10194

to the mapping function (5) while the non-linear parameter a_1 and 11 ($N = 10$) linear expansion coefficients, b_k, determine the $d^{se}(r)$ function (4) itself.

At equilibrium, $r_e^{BO} = 1.12822960$ Å for the BO potential (1.12821704 Å for the full potential including mass-dependent corrections) and $d^{se}\left(r_e^{BO}\right) = -0.1224$ D. The dissociation energy is $D_e = 90679.1$ cm^{-1} (independent of the mass corrections). The expectation value of the DMF in the ground state, $\langle 00|d^{se}(r)|00\rangle = -0.1098$ D, coincides with experiments within the experimental uncertainties [47,49,50].

The discrete variation of the integer parameter n has demonstrated that the minimal set of the Chebyshev expansion parameters, b_k, corresponds to $n = 2$ while the optimal value of a_0 should be close to $n/r_e \approx 1.77$, cf. Table 3. The short-range, $A_3^{se} \approx +920$ DÅ^{-3}, and long-range, $d_4^{se} \approx -5.0$ DÅ^4, coefficients calculated with the fitted a_1 and b_k parameters according to relations (7) were found to be rather close to their ab initio pure atomic counterparts, $A_3 = +1008$ DÅ^{-3} [35] and $d_4 = -5.1 \pm 0.1$ DÅ^4 [37], respectively (see details in the supplementary file **d4_data_analysis.pdf**).

The ab initio DMF points obtained in the framework of the MR-ACPF method as expectation values in the long-range region $r > 5.5$ Å (see the Inset (b) on Fig. 2) and by the MR-ACPF/FF approximation for the intermediate region $r \in [0.5, 5.5]$ Å were involved in the regularization NLLSF procedure (3). The corresponding CCSD(T)/FF values localized on the interval $r \in [0.5, 1.6]$ Å were also included in the NLLSF procedure. The required uncertainties in the ab initio DMF points, σ_i^{ab}, were estimated as the difference of the DMF results obtained with the different atomic basis sets and alternative electron-correlation account methods. It should be noted that the resulting σ_i^{ab} values are essentially less than the difference between $d^{CCSD(T)}(r)$ and $d^{ACPF}(r)$ functions.

To propagate smoothly the $d^{se}(r)$ function (4) from $r = 0.5$ Å to the UA limit, six external ab initio points, evaluated under the UA approximation at extremely short $r < 0.032$ Å in Ref. [35], were included in the NLLSF procedure (3) as well. The particular behavior of the DMF at small distances ($r < 0.5$ Å) should not apparently affect the majority part of the calculated TDMs, except a few "abnormal" and extremely weak transitions.

The derived semi-empirical DMF reproduces the most part of the measured intensities within their experimental uncertainty (see Table 2).

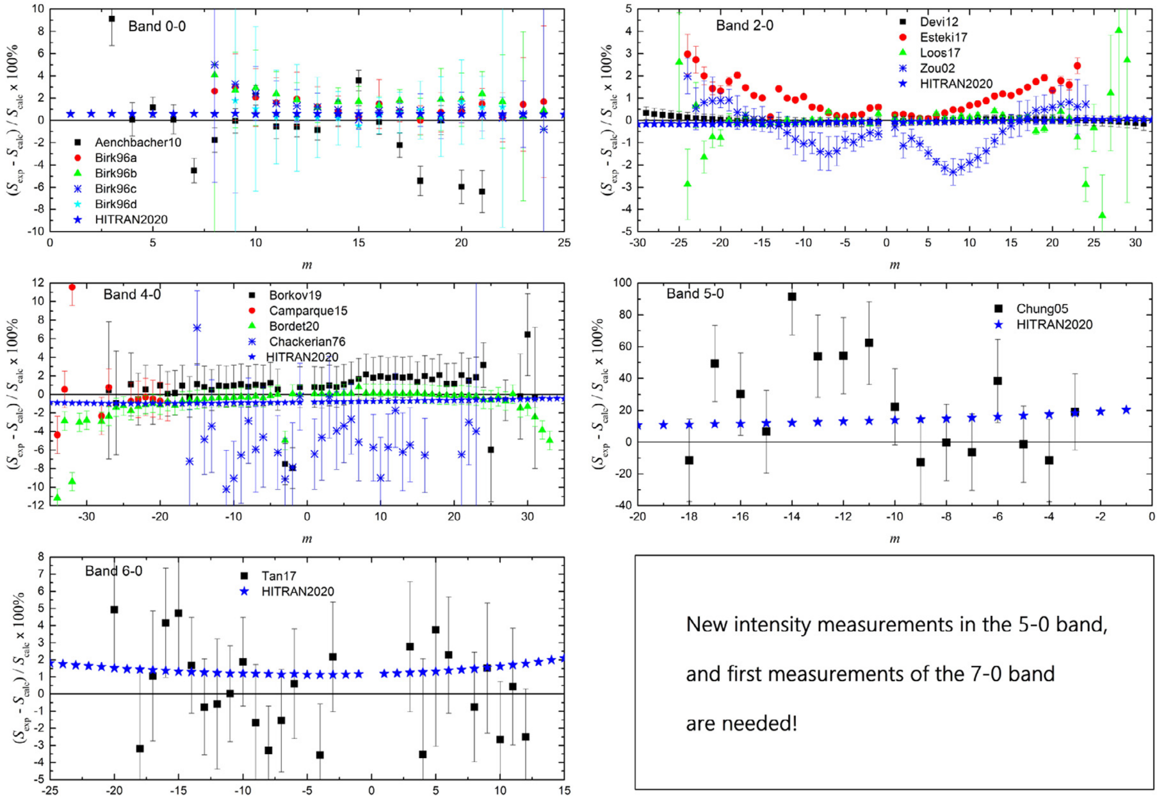

Fig. 4. Comparison of line intensities of $^{12}C^{16}O$ 0-0, 2-0, 4-0, 5-0 and 6-0 bands with the present semi-empirical study as a reference. Note the different scales in the panels of the plot. "Calc" refers to semi-empirical results from the present study, while "exp" refers to the experimental studies from Refs. [50,52] for the 0-0 band, Refs. [23,55] for the 2-0 band, Refs. [25,28,61,62] for the 4-0 band, Ref. [63] for the 5-0 band, Ref. [64] for the 6-0 band or HITRAN2020 [30]. Note that for HITRAN2020 intensities for these bands are same as in Li et al. [12]/HITRAN2016 [4].

3.4. The new line list

Using the PEF from Meshkov et al. [16] and the semi-empirical DMF described here, we calculated new line lists for six stable ($^{12}C^{16}O$, $^{13}C^{16}O$, $^{12}C^{18}O$, $^{12}C^{17}O$, $^{13}C^{18}O$, $^{13}C^{17}O$, and three radioactive isotopologues ($^{14}C^{16}O$,$^{14}C^{18}O$, $^{14}C^{17}O$) of CO. We already discussed that the new PEF allows us to calculate more accurate line positions (see Fig. 1, for example). In this section, we will show that the intensities are also superior with respect to the previous efforts. It is compared to the selected experimental works for different bands. We chose the comparison to parallel the way it was carried out on Fig. 6 of Li et al. [12]. The comparisons are shown in Figs. 3 and 4 for seven bands of $^{12}C^{16}O$ (from pure rotational to fifth overtone bands). Noticeable differences with the intensities calculated in Li et al. [12] can be observed for the fundamental and second overtone bands, provided in Fig. 3. This is because new high-quality experimental data from Refs. [26,27,29] were used as input in this work. It is worth reminding that for HITRAN2020 edition [30] the intensities of these bands from Li et al. were scaled to agree better with these new experiments. This, however, is a much less efficient approach than the one used here, especially for the hot bands with Δv=1 and 3. Particularly interesting in the 3-0 band is the rotational dependence of the deviation of the current work and Li et al. (used in HITRAN2016 [4] for all the bands) and consequently HITRAN2020. The skewed dependence can be explained by the fact that Li et al. used the data from Ref. [57] (with reported very low uncertainty) in their fit. Considering that the uncertainties of these two P-lines were much lower than other works used in the fit for this band, the intensities in Li et al. [12] bent to match these values. However, the authors of Ref. [57] have later identified [27] that the intensities in their original work were underestimated by a few percent. In HITRAN2020 the intensities from Li et al. were scaled up to 2.6% to match the measured value of a single relatively high-*J* R transition, and therefore, although intensities have come much closer to their current values (especially for the R-lines), intensities of the *P*-branch lines were still too low, although much better than in Li et al./HITRAN2016. The rest of the bands are shown in Fig. 4. It is not surprising that for most of the bands, the agreement with HITRAN2020 (same as Li et al. and HITRAN2016 for these bands) is much closer, as much of the same data was used in the input. Nevertheless, about 1% deviation can be observed for the third and fifth overtones, likely due to the fact that in their fit Li et al. used only preliminary and smaller dataset from experiments described in Refs. [25,61,64]. The differences for the 5-0 band are clear and are due to very large experimental uncertainties [63], the difference in the ab initio calculations, and interpolation techniques for the DMF points. Indeed, in Ref. [63] a few P-lines were measured under very high pressure and low signal-to-noise ratio. These measurements will need to be revisited with sensitive techniques.

3.5. The issue of isotopic abundances

We checked if the new dipole moment function eliminates the discrepancies observed in Li et al. [12] when comparing calculated intensities of the isotopologues with those measured in different bands. These discrepancies were mentioned in the Introduction, and the question remained if the issues were related to the measurements or calculations. Unfortunately, it appears that these discrepancies still exist. Nevertheless, with the availability of new low-uncertainty measurements in different bands, it is now easier to attribute the issue to the relative isotopic abun-

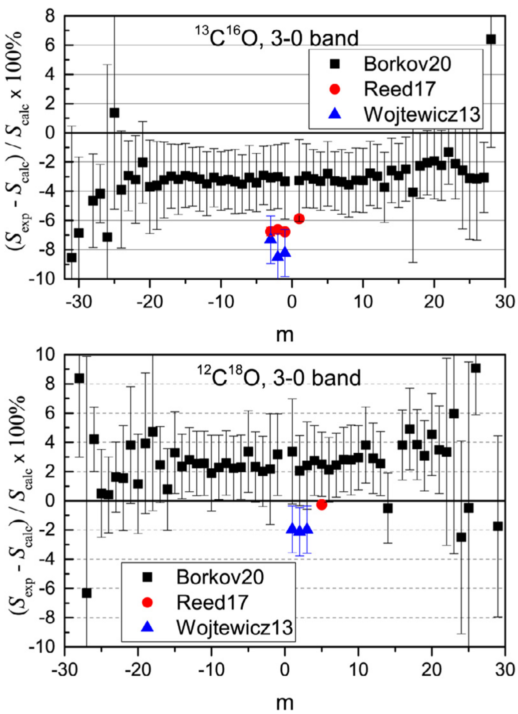

Fig. 5. Comparison of the intensities calculated in this work using the semi-empirical DMF and those measured in Refs. [24,29,57] for $^{13}C^{16}O$ (top panel) and $^{12}C^{18}O$ (bottom panel).

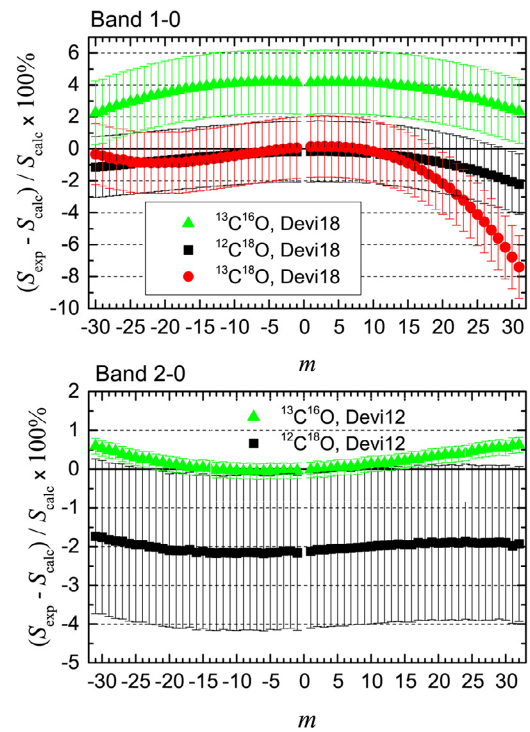

Fig. 6. Comparison of the isotopologue intensities calculated in this work using the semi-empirical DMF and those measured in Refs. [26,68] for the fundamental band (top panel) and the first overtone (bottom panel).

dances in experimental cells. The second overtone is the only band where the highly accurate experimental measurements of the intensities of minor isotopologues were reported in more than two works. Fig. 5 shows comparisons of the intensities calculated in this work and those measured with FTS in Tomsk [24] and CRDS in Torun [57] and NIST [29]. It is clear that the discrepancies for the $^{13}C^{16}O$ and $^{12}C^{18}O$ are systematically different compared to various experiments and do not overlap within their uncertainty boundaries. With that, we should remember that intensities are calculated here for principal isotopologue and those measured in Refs. [24,29] agree really well. At the same time, the intensities of the principal isotopologue measured in Ref. [57] do not agree for the reasons discussed in the previous section. In that regard, the uncertainties of the isotopologues intensities in Ref. [57] should probably be considered larger than the ones reported. Similar situation is observed for other isotopologues and in other bands. The only issue is that for the least abundant isotopologues the uncertainties of the measurements are much higher. While CRDS experiments are still mostly systematically different, those obtained in “natural” abundance with FTS have random and some times extremely high (over hundred percent) deviations. Interestingly, the authors of Refs. [25,61] have also highlighted systematic deviation of the HITRAN2016 intensities in comparison with their CRDS intensities in the 4-0 band of the $^{13}C^{16}O$ isotopologue [25]. They did attribute this to a different isotopic constitution of their sample from the one assumed in HITRAN. This is in accord with our conclusions. An overview plot with all isotopologues in bands 3-0 and 4-0 is given in the supplementary file **Isotopologues.pdf**.

It is also interesting to consider high-quality measurements of the fundamental and first overtone bands reported by Devi et al. [26,68]. These measurements did include isotopologues, furthermore, experiments carried out for the first overtone featured enriched samples [68]. Fig. 6 shows these comparisons. It is clear that for both bands, intensities do fall within their experimental uncertainties, except for the 1-0 band of $^{13}C^{16}O$. However, for the 2-0 band, when the $^{13}C^{16}O$ sample was enriched, the agreement was perfect. The smoothly increasing deviations for the high-J transitions of the $^{13}C^{18}O$ isotopologue in the 1-0 band could be explained by the fact that in Ref. [26] multispectrum fit approach was fitting directly to Herman–Wallis polynomial for the entire band, rather than individual intensities. Considering low signal-to-noise ratio of the high-J transitions of this minor isotopologue, it can be assumed that including these transitions into the fit have slightly obscured the polynomial obtained in the experiment.

These observations do give us confidence that the intensities calculated using our semi-empirical DMF, which was fit only to the selected measurements of the principal isotopologue of CO can accurately predict intensities for the isotopologues. Existing discrepancies with experimental measurements should be attributed to the uncertainties in the abundances in experimental samples. We would like again to suggest that the involvement of the mass spectrometers would allow resolving these issues.

4. Summary

In this work, the mass-invariant permanent electric dipole moment function has been reconstructed for the ground state of the CO molecule in the entire range of the inter-nuclear distance by means of the simultaneous least-squares fitting of the experimental intensity distributions currently available for the main $^{12}C^{16}O$ isotopologue and the ab initio point-wise dipole moment functions evaluated using alternative quantum-chemistry methods in a wide range of r. The high-precision (0.07%) experimental 3-0 R(23) line intensity [27] was used in the fitting, and the model DMF reproduced the intensity with the 0.04% accuracy. Additionally, the model DMF was restricted to obey the theoretical behavior in the limits of small and large r. Such an involved procedure was applied for the first time for the dipole moment functions to the best of our knowledge.

The input experimental intensity data were selected from both the early studies and recent investigations. The analytical forms invented for the DMF approximation were constrained to possess the physically justified asymptotic behavior in both the united atom and dissociation limits. New line lists were generated for all of the CO isotopologues (including those containing ^{14}C), and the issues described in the Introduction are effectively attended. In contrast to the line positions, the regular Born–Oppenheimer breakdown provides negligible contribution into the semi-empirical DMF and TDM-values of the CO ground state at least for the lowest vibrational levels with $J < 50$.

It was shown that the relative intensities of the isotopologue transitions are calculated accurately, and the discrepancies observed in the literature are linked to the isotopic abundances in experimental cells that do not coincide with "natural" abundances assumed in HITRAN. This is a significant result for the applications. For instance, measurements of the isotopic makeup of the atmospheric CO can assist in the determination of its various sources and sinks [1]. Precise knowledge of the relative intensities of CO isotopologues are also important for the determination of carbon and oxygen isotopic fractionations in solar [8] and stellar [9] atmospheres.

We propose that the results of this work should be considered for the future updates of HITRAN and HITEMP line lists. It would be trivial to supplement it with the broadening parameters using semi-empirical models explained in Hashemi et al. [69]. At the moment the line list is limited to $\Delta v = 6$ with $v \le 41$, because that is where we are confident in its quality. Nevertheless, the high-quality ab initio calculations of the DMF carried out in this work provide for correct behavior of the DMF in the limits of small and large inter-atomic separations, which will help extend the calculations towards the high-overtone transitions. This nontrivial task will be carried out in the follow up investigations.

The semi-empirical matrix elements should not ideally depend on any particular form (analytical or numerical) selected for the DMF approximation. However, in practice it is not true at least for very small TDM values corresponding, for instance, to "abnormal" transitions [18] and/or very high overtones [70]. In particular, the conventional "natural" cubic- and quintic-spline interpolation of the ab initio DMF points, is not suitable to describe the high overtone intensities. Furthermore, including the high overtones in the fit indispensably requires to use a model DMF having high continuous derivatives with respect to r.

To monitor this "approximation effect", an alternative, so-called "irregular", analytical form of the semi-empirical DMF function has been recently developed in Refs. [20–22]. The implementation of the "irregular" function provides for the almost equivalent description of the experimental data currently available for all CO isotopologues, except the "abnormal" 5-0 band [71]. Therefore, refined measurements of the rotational intensity distribution in the 5-0 band and/or in higher overtones, $\Delta v > 6$, is crucially important now from both experimental and theoretical viewpoints.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Vladimir V. Meshkov: Investigation, Formal analysis, Methodology, Validation. **Aleksander Yu. Ermilov:** Investigation. **Andrey V. Stolyarov:** Conceptualization, Project administration, Writing – review & editing, Writing – original draft. **Emile S. Medvedev:** Project administration, Writing – review & editing. **Vladimir G. Ushakov:** Formal analysis, Methodology, Validation, Visualization. **Iouli E. Gordon:** Conceptualization, Project administration, Writing – review & editing, Writing – original draft.

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Appendix A. The mapping function for the radial coordinate

For optimal representation of the semi-empirical DMF in the entire range of internuclear distance by means of a minimal number of the adjustable fitting parameters we are looking for a new class of analytical functions which create the proper density of the r-grid points in the region where wavefunctions of the vibrational levels of interest are mainly localized. For the lowest vibrational states of CO under consideration it is obvious the region in a vicinity of the equilibrium distance r_e of the ground state.

To perform the $r \to z$ coordinate transformation unambiguously the mapping function $z(r)$ must monotonically increase as r increases, and, hence, its density function $\rho(r) = dz/dr > 0$ is always positive defined. Here, we introduce a class of the two-parametric mapping functions having the common density function

$$\rho(r) \propto r^n e^{-\alpha r} \tag{A.1}$$

where $n > 0$ is integer number while $\alpha > 0$ is the real one. The $\rho(r)$ function (A.1) has a single maximum near the point $r = n/\alpha$, and its integration under the boundary condition $z(0) = -1$ and $z(\infty) = 1$ yields the mapping function

$$z(r) = 1 - 2e^{-\alpha r} \sum_{k=0}^{n} \frac{(\alpha r)^k}{k!} \tag{A.2}$$

which is very similar to Tang–Toennies damping function [72] basically used for the correct representation of the asymptotic long-range tail of interatomic potentials. The invented analytical mapping function (A.2) would be also appropriated to increase efficiency a numerical solving of the radial Schrodinger Eq. (9) using both finite-difference and collocation methods [73].

Supplementary material

File **Readme_Meshkov21.pdf** (mmc1.zip). Explanation of the column headings in some files.

File **LineList_Regular_dv6.rar** (mmc2.zip). Line lists of all nine CO isotopologues calculated with the regular DMF of Eq. (4) up to the fifth overtone inclusive.

File **CO_DM_out.txt** (mmc3.zip). The database with TDMs calculated with the regular DMF of Eq. (4).

File **CO_DM.dat** (mmc4.zip). The values of the regular DMF of Eq. (4) in the range of $r = 0.001$–15 Å.

File **CO_DM.f** (mmc5.zip). FORTRAN code for the regular DMF of Eq. (4) with 13 adjustable parameters.

File **Present_study_v1.rar**(mmc6.zip). The ab initio results of the present study. It contains the following files.

File **CO_X_CCSD(T).dat**. The ab initio DMF of the present study. The CCSD(T) method with the cc-pV5Z basis set. Only $r = 0.5$–1.8 Å is used in the fitting.

File **PS_MP2_RHF.txt**. The ab initio DMF of the present study. The MP2 and RHF methods, $r = 0.11$–1.25 Å. Not used in the fitting.

File **PS_MRDCI.txt**. The ab initio DMF of the present study. The MRDCI with the cc-pVQZ and aug-cc-pVQZ basis sets, $r =$ 1.1–15

Å. Not used in the fitting. Used to estimate the uncertainty of the ACPF data.

File **CO_X_DM_ACPF_f_pwcV5Z_sr.dat**. The ab initio DMF of the present study. The ACPF method with the pwcV5Z basis set, $r <$ 0.5 Å.

File **CO_X_DM_ACPF.dat**, the same as above, $r = 0.5$–17 Å.

File **Buldakov09.rar** (mmc7.zip). The ab initio values of the CO dipole-moment function in the united-atom limit.

File **Li15_TDM_recalculated_v1.rar** (mmc8.zip). The TDM values from Table S3 of **Li15** with recalculated uncertainties as explained in the **Readme_Li15_ recalculated.txt** file.

File **Li15_DMF.rar** (mmc9.zip). The ab initio DMF of **Li15** supplemented with the DMF uncertainties.

File **Q.rar** (mmc10.zip). Statistical sums for all isotopologues calculated with the **Meshkov18** potential at $T = 1$–9000 K. Column 1, temperature in K; column 2, statistical sum without nuclear weights; column 3, statistical sums with nuclear weights.

File **d4_data_analysis.pdf** (mmc11.zip). Precise determination of the long-range coefficient.

File **Isotopologues.pdf** (mmc12.zip). The figure showing the relative differences between the calculated and measured intensities in the 3-0 and 4-0 bands of isotopologues.

Supplementary material associated with this article can be found, in the online version, at 10.1016/j.jqsrt.2022.108090

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